C.A.N.I.S.
Corrosion And Nuclear Inspection System

Drum Store Inspection Report

Drum Store 1 (simulated) | 2026-11-08

Store	Drum Store 1 (simulated)	Report number	CANIS-preview8_014613
Inspected on	2026-11-08 01:46:26 UTC	Time taken	1 min 30 s
Inspected by	C.A.N.I.S. inspection robot	Status	Practice run in a simulated store

Summary

Drums checked	**8 of 8**	Areas safe to enter	**1**
• Need action	**6**	Areas restricted	**4**
• Check again	**1**	Areas to keep out of	**0**
• No problem found	**1**	Areas not checked	**7**
Radiation dose to the robot	**27% of its limit**	Time taken	**1 min 30 s**

6 drums need action and 1 should be checked again. 4 areas are restricted. 7 areas were not checked.

How to read this report

- **Action needed.** Damage seen clearly. An engineer should assess the drum.
- **Check again.** Possible damage, not yet confirmed.
- **No problem found** in this inspection.

Safe	Normal access.
Restricted	Enter only under controlled-area rules.
Keep out	Do not enter.
Not checked	The robot did not measure here: treat as unknown.

Radiation is given in microsieverts per hour (µSv/h). Above 7.5 µSv/h an area needs controlled-area rules; above 100 µSv/h people should keep out.

1 Who can go where

The robot measured radiation as it walked. Each aisle is split into 2-metre sections, and each section is given one of four ratings. A section is only marked safe if the robot actually measured it.

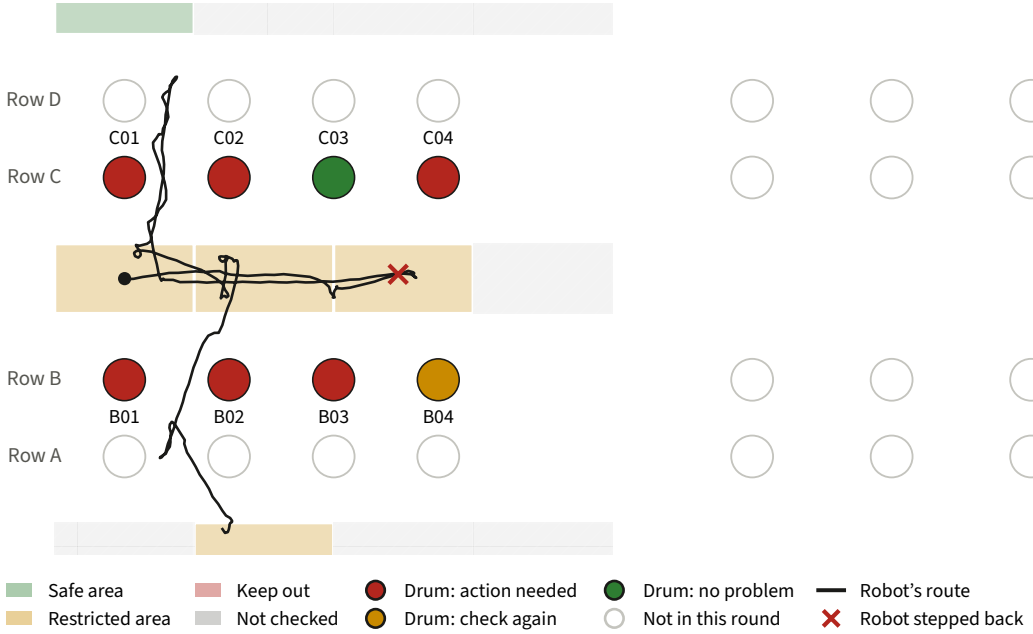


Figure 1. Plan of the store. Coloured bands are the aisle sections; circles are drums.

Where	Section	Rating	Why
Walkway beside row A	2–4 m	Restricted	Radiation reached 8 µSv/h, above the controlled-area level.
Aisle between rows B and C	0–2 m	Restricted	Damaged drum here: B01, C01.
Aisle between rows B and C	2–4 m	Restricted	Radiation reached 70 µSv/h, above the controlled-area level.
Aisle between rows B and C	4–6 m	Restricted	Radiation reached 79 µSv/h, above the controlled-area level.
Walkway beside row D	0–2 m	Safe	Radiation stayed normal (highest 5.7 µSv/h).
Walkway beside row A	0–2 m	Not checked	The robot did not go here this time.
Walkway beside row A	4–6 m	Not checked	The robot did not go here this time.
Walkway beside row A	6–8 m	Not checked	The robot did not go here this time.
Aisle between rows B and C	6–8 m	Not checked	The robot did not go here this time.
Walkway beside row D	2–4 m	Not checked	The robot did not go here this time.
Walkway beside row D	4–6 m	Not checked	The robot did not go here this time.
Walkway beside row D	6–8 m	Not checked	The robot did not go here this time.

Sections are measured along the aisle from the end nearest drum 1. Hydrogen gas was not measured in this inspection.

2 Every drum at a glance

All drums in this inspection, most urgent first.

Drum	Waste	Where	Result	What was seen	How sure	Radiation
B01	Low-level	Row B, drum 1	Action needed	Rust and bulging	Low (49%)	Raised (7.7 µSv/h)
B02	Low-level	Row B, drum 2	Action needed	Bulging	Low (42%)	Raised (25 µSv/h)
B03	Low-level	Row B, drum 3	Action needed	Rust and bulging	High (88%)	Raised (77 µSv/h)
C01	Low-level	Row C, drum 1	Action needed	Rust and bulging	High (83%)	Raised (8.7 µSv/h)
C02	Low-level	Row C, drum 2	Action needed	Rust	High (82%)	Normal (7.0 µSv/h)
C04	Low-level	Row C, drum 4	Action needed	Rust	High (74%)	Raised (79 µSv/h)
B04	Low-level	Row B, drum 4	Check again	Bulging	Low (36%)	Raised (78 µSv/h)
C03	Low-level	Row C, drum 3	No problem found	Nothing found	–	Raised (77 µSv/h)

“How sure” is how confident the camera software was about the damage it saw. This is the first inspection of these drums, so there is nothing to compare with yet. Not measured in this inspection: drum surface temperature, hydrogen gas, surface wipe (swab) results. The robot could not read the drum labels, so drums were identified by their position in the store.

3 Drums that need attention

Drum B01

Action needed

Low-level waste | Row B, drum 1

No photo was saved for this drum.

What was seen	Rust and bulging
How sure	Low (49%)
Why this rating	Rust and bulging seen clearly from more than one angle.
Radiation at drum	Raised (7.7 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	Not measured

What to do: Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).

Drum B02

Action needed

Low-level waste | Row B, drum 2

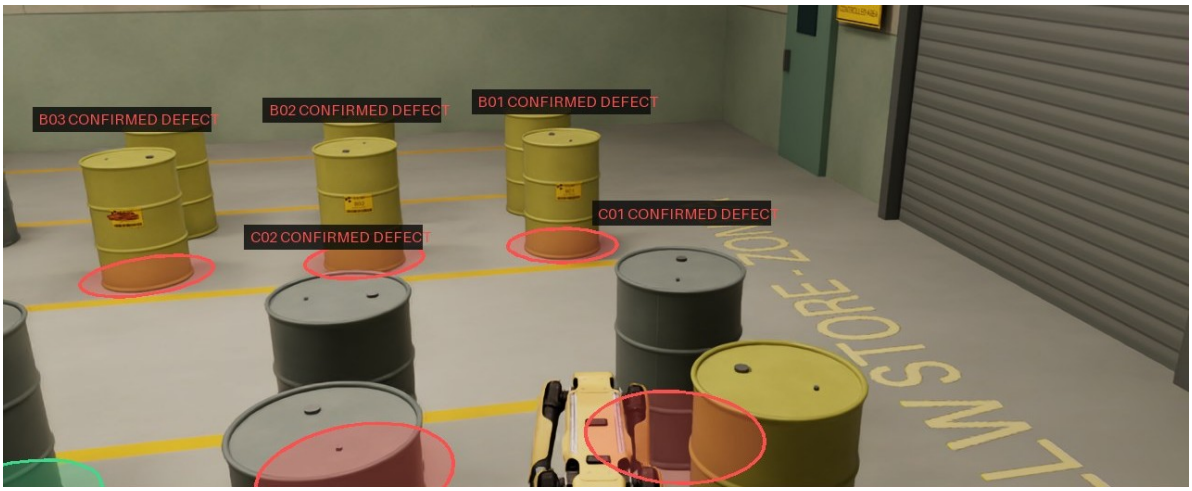


Photo taken 0:39 into the inspection. The ring on the floor marks drum B02.

What was seen	Bulging
How sure	Low (42%)
Why this rating	Bulging seen clearly from more than one angle.
Radiation at drum	Raised (25 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	Not measured

What to do: Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).

Drum B03

Action needed

Low-level waste | Row B, drum 3

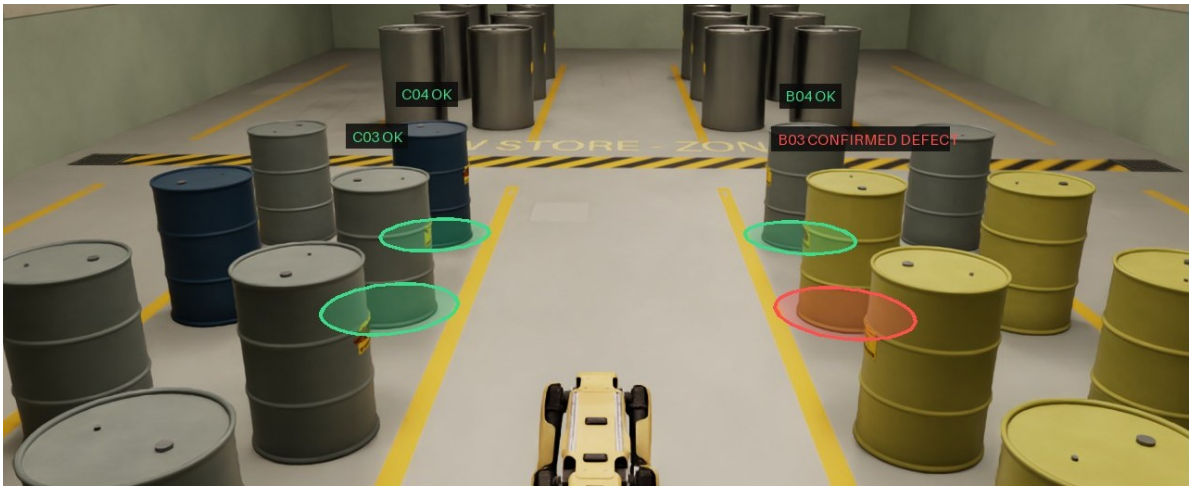


Photo taken 0:01 into the inspection. The ring on the floor marks drum B03.

What was seen	Rust and bulging
How sure	High (88%)
Why this rating	Rust and bulging seen clearly from more than one angle.
Radiation at drum	Raised (77 µSv/h)
Surface wipe (swab)	Taken, but the result was not recorded
Surface temperature	Not measured

What to do: Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).

Drum C01

Action needed

Low-level waste | Row C, drum 1

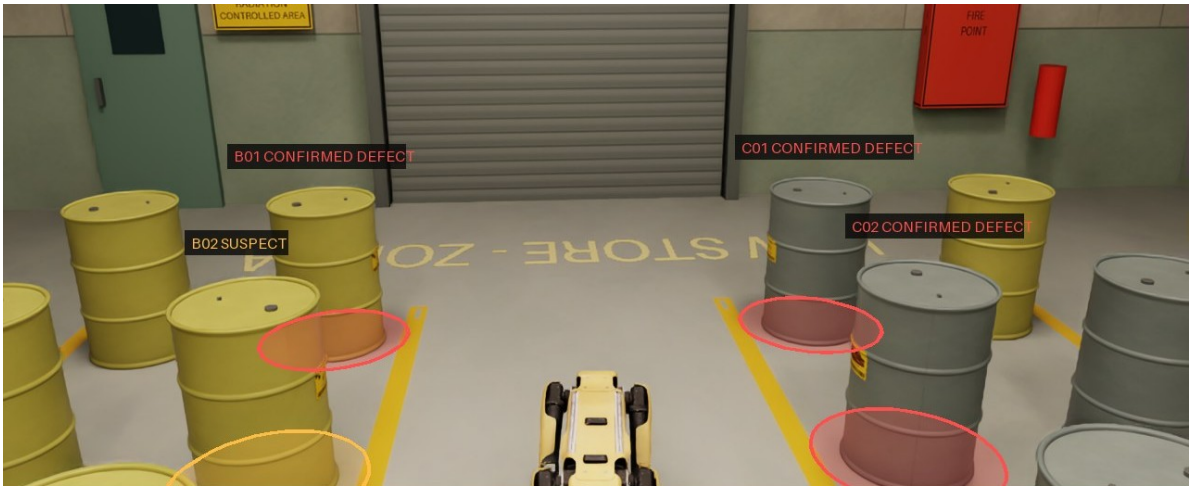


Photo taken 0:23 into the inspection. The ring on the floor marks drum C01.

What was seen	Rust and bulging
How sure	High (83%)
Why this rating	Rust and bulging seen clearly from more than one angle.
Radiation at drum	Raised (8.7 µSv/h)
Surface wipe (swab)	Taken, but the result was not recorded
Surface temperature	Not measured

What to do: Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).

Drum C02

Action needed

Low-level waste | Row C, drum 2

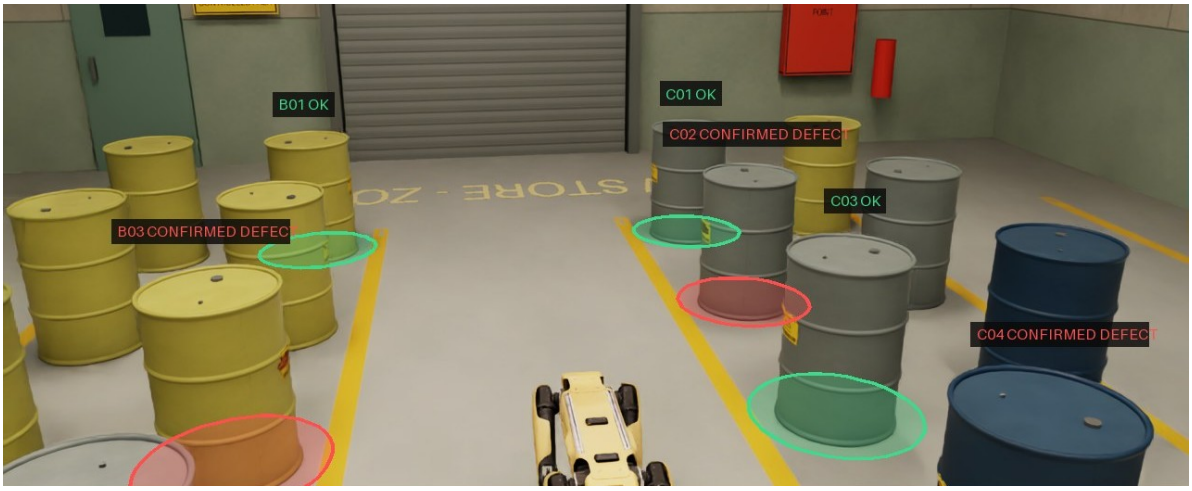


Photo taken 0:19 into the inspection. The ring on the floor marks drum C02.

What was seen	Rust
How sure	High (82%)
Why this rating	Rust seen clearly from more than one angle.
Radiation at drum	Normal (7.0 µSv/h)
Surface wipe (swab)	Taken, but the result was not recorded
Surface temperature	Not measured

What to do: Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

Drum C04

Action needed

Low-level waste | Row C, drum 4

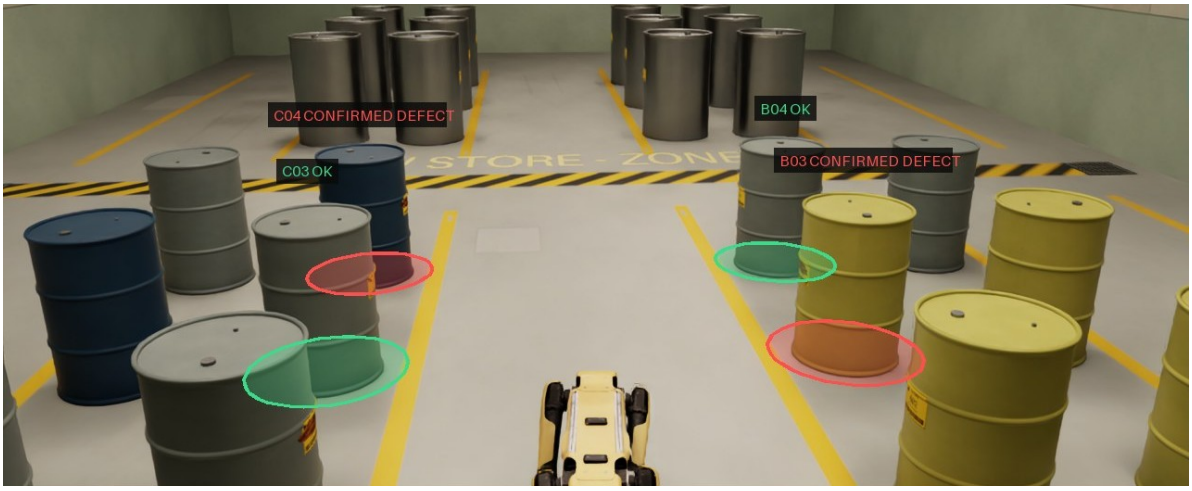


Photo taken 0:03 into the inspection. The ring on the floor marks drum C04.

What was seen	Rust
How sure	High (74%)
Why this rating	Rust seen clearly from more than one angle.
Radiation at drum	Raised (79 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	Not measured

What to do: Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

Drum B04

Check again

Low-level waste | Row B, drum 4

No photo was saved for this drum.

What was seen	Bulging
How sure	Low (36%)
Why this rating	Possible bulging, seen from only one angle.
Radiation at drum	Raised (78 µSv/h)
Surface wipe (swab)	Not taken
Surface temperature	Not measured

What to do: Inspect again within 7 days, with a closer look and a surface wipe (swab).

4 What was not checked

- Walkway beside row A, sections 0–2 m, 4–6 m, 6–8 m: the robot did not walk here this time, so these are unknown.
- Aisle between rows B and C, sections 6–8 m: the robot did not walk here this time, so these are unknown.
- Walkway beside row D, sections 2–4 m, 4–6 m, 6–8 m: the robot did not walk here this time, so these are unknown.

Not measured anywhere in this inspection: drum surface temperature, hydrogen gas, surface wipe (swab) results.

Radiation event. 0:15 into the inspection, radiation in the aisle rose from about 7 to 41 µSv/h. The robot stepped back to a safer spot, waited for it to settle, and returned 0:26 into the inspection to finish the drums it had skipped.

5 How accurate was the robot?

This was a practice run in a simulated store, so we know which drums were really damaged. The robot was not told; the comparison is made only for this report.

- It found 3 of the 3 damaged drums.
- It also raised 4 false alarms: B01, B02, B04, C01 were actually undamaged.

A false alarm costs a person an extra check. A missed drum is the more serious error.

6 How this report was made

The robot walks the aisles with a camera and a radiation meter. Software trained to recognise rust and bulging (swelling) on drums points out possible damage in the camera images. When it is unsure, the robot goes back for a closer look from another angle. Fixed safety rules, the same every time, then decide each rating in this report. An AI assistant may add a short written explanation for drums that need action, but it cannot change any rating or number. Everything in this report comes from what the robot recorded during the inspection.

Limits used

What	Level	Source
Restricted area	above 7.5 µSv/h	HSE, *Work with ionising radiation* (L121), regulation 17. Used here as an on-the-spot reading.
Keep out	above 100 µSv/h	HSE, L121 guidance on high instantaneous dose rates.
Surface wipe	above 0.4 / 4 Bq/cm^2	IAEA, *Regulations for the Safe Transport of Radioactive Material* (SSR-6), para. 508.
Warm drum	5 °C above the air	Our own rule for this project; not an official limit.
Hydrogen gas	no level set yet	We have not yet found an official alarm level to cite, so hydrogen does not change any rating.

Please note. This inspection took place in a computer-simulated store, not a real one. Radiation values come from a simplified model. The damage-spotting software was trained on computer-generated images. How far rust has spread is estimated from photos, not measured. Only a surface wipe can show contamination. The format follows the themes of the Nuclear Decommissioning Authority's *Guidance on Monitoring of Waste Packages During Storage* (WPS/640).

A Radiation during the inspection

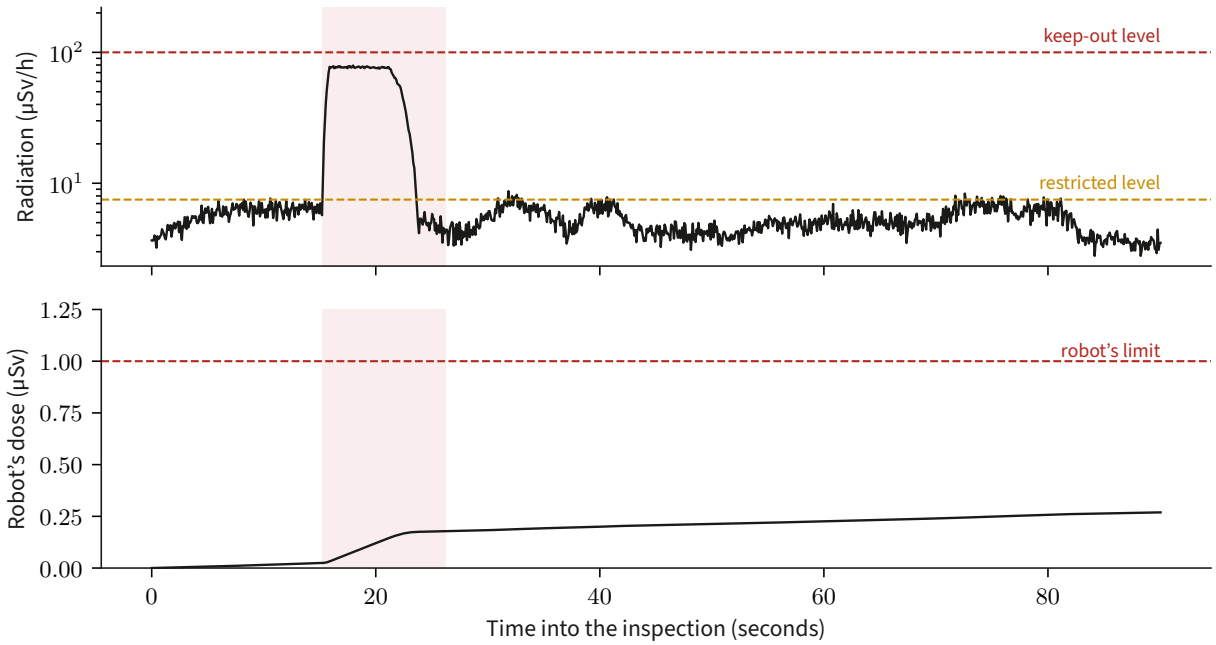


Figure 2. Top: radiation measured by the robot as it walked (dashed lines are the restricted and keep-out levels). Bottom: the total dose the robot received, against its limit. The shaded band is the radiation event.

B Step-by-step record

What the robot did and why, in order. Times are minutes:seconds into the inspection.

Time	Drum	What happened
0:00	B03	Possible rust on B03. Went back for a second look.
0:00	B03	Rust on B03 confirmed from a second angle.
0:02	C04	Possible rust on C04. Went back for a second look.
0:02	C04	Rust on C04 confirmed from a second angle.
0:05	B03	Went to drum B03 for a closer inspection.
0:07	B03	Took a surface wipe (swab) at drum B03.
0:14	C04	Went to drum C04 for a closer inspection.
0:16	C04	Radiation rose sharply. The robot stepped back to a safer spot to wait.
0:18	C02	Possible rust on C02. Went back for a second look.
0:19	C02	Rust on C02 confirmed from a second angle.
0:20	B01	Possible rust on B01. Went back for a second look.
0:21	B01	Rust on B01 confirmed from a second angle.
0:21	B02	Possible bulging on B02. Went back for a second look.
0:22	C01	Possible rust on C01. Went back for a second look.
0:22	C01	Rust on C01 confirmed from a second angle.
0:26		Radiation back to normal. The robot carried on with the round.
0:34	B01	Went to drum B01 for a closer inspection.
0:38	B02	Bulging on B02 confirmed from a second angle.
0:43	C01	Went to drum C01 for a closer inspection.

Time	Drum	What happened
0:46	C01	Took a surface wipe (swab) at drum C01.
0:50		Carried on with the round.
0:53	B02	Went to drum B02 for a closer inspection.
0:56	C02	Went to drum C02 for a closer inspection.
1:01	C02	Took a surface wipe (swab) at drum C02.
1:05		Carried on with the round.
1:18	B04	Possible bulging on B04. Went back for a second look.

The robot also looked at drums A01, A02, A03, A04, D01 on its way. They are outside this inspection and not reported here.

C Technical details

For engineers and auditors.

Run ID	`preview8_014613`
Software version	`a1ff9ee`
Damage detector	YOLO26s, run `yolo26s_20261107T225757Z_d625ef`, reporting cut-off 0.25
Written explanations	NVIDIA Cosmos Reason 2B (drums needing action only; checked against the recorded facts)
Note	The explanation service was unavailable during this run, so the fixed rules made every decision.
Simulator	NVIDIA Isaac Sim
Status names	Action needed = `Confirmed defect`; Check again = `Suspect`; No problem found = `OK`; Not inspected = `Couldn't inspect`